

HOPETOUN WA, Australia

WMO: 956350

Lat: 33.931S

Lon: 120.128E

Elev: 27

StdP: 101.00

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	5.9	-0.9	3.5	18.3	1.1	4.1	14.2	12.9	13.7	11.4	14.2	3.3	40	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.5	31.4	18.0	28.1	17.7	25.6	17.6	20.9	25.0	20.2	23.7	19.6	22.8	6.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	14.6	21.7	19.1	13.9	21.2	18.5	13.4	21.0	60.6	25.1	58.1	23.7	56.1	22.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.2	9.2	2.5	38.7	1.2	2.3	1.6	40.3	0.9	41.7	0.3	42.9	-0.6	44.6		
(5)			1.8	22.8	1.1	1.0	1.0	23.6	0.3	24.2	-0.3	24.7	-1.1	25.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	19.2	19.5	18.9	17.6	15.4	13.5	12.6	12.9	13.7	15.1	16.8	17.9	(6)
(7)		DBStd	3.39	2.19	2.32	2.65	2.64	2.43	1.97	1.95	2.14	2.38	2.69	2.58	2.28	(7)
(8)		HDD10.0	5	0	0	0	0	0	1	2	1	1	0	0	0	(8)
(9)		HDD18.3	1018	12	10	23	44	97	147	178	168	139	107	59	33	(9)
(10)		CDD10.0	2219	285	265	274	227	167	104	83	93	113	159	205	244	(10)
(11)		CDD18.3	192	39	42	39	21	6	1	0	1	2	8	14	19	(11)
(12)		CDH23.3	1155	143	159	223	149	55	2	1	17	40	118	131	117	(12)
(13)		CDH26.7	436	54	57	96	49	10	0	0	3	11	50	57	49	(13)
(14)	Wind	WSAvg	4.8	5.4	5.3	5.0	4.4	4.2	4.4	4.6	4.9	4.9	5.3	5.2	(14)	
(15)	Precipitation	PrecAvg	519	27	22	27	40	69	61	63	62	53	42	33	18	(15)
(16)		PrecMax	764	161	68	109	108	330	144	153	124	113	170	150	79	(16)
(17)		PrecMin	282	0	0	1	5	7	17	10	14	4	0	0	0	(17)
(18)		PrecStd	99	33	19	23	25	57	32	29	28	27	30	28	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	34.0	35.4	31.4	28.0	23.2	22.0	25.4	28.1	33.3	33.3	32.6	(19)
(20)		0.4%	MCWB	19.7	20.4	19.6	17.2	16.1	13.7	13.0	13.7	14.5	16.0	16.8	17.9	(20)
(21)		2%	DB	26.8	27.3	28.9	27.5	24.6	20.9	19.9	21.7	23.5	26.5	26.6	26.1	(21)
(22)		2%	MCWB	19.8	19.3	18.9	17.1	15.5	13.1	12.9	13.6	14.4	15.6	16.8	18.6	(22)
(23)		5%	DB	24.1	24.6	25.0	24.3	22.2	19.4	18.3	19.5	21.0	22.2	23.0	23.2	(23)
(24)		5%	MCWB	19.1	19.3	18.5	16.8	15.1	13.3	12.7	13.2	13.9	15.4	17.0	18.0	(24)
(25)		10%	DB	22.8	23.1	23.0	22.2	20.2	18.0	17.1	17.8	19.0	19.7	21.0	21.9	(25)
(26)		10%	MCWB	18.3	18.7	18.3	17.1	14.7	13.0	12.1	12.7	13.4	14.9	16.4	17.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	22.1	21.5	20.1	18.6	16.8	15.5	16.2	16.9	18.2	19.6	20.6	(27)
(28)		0.4%	MCDWB	27.5	26.3	25.4	23.5	23.3	19.0	18.9	20.8	23.5	23.2	24.5	26.2	(28)
(29)		2%	WB	20.7	20.8	20.4	19.2	17.2	15.3	14.2	15.0	15.6	17.1	18.6	19.4	(29)
(30)		2%	MCDWB	24.8	24.3	24.1	22.3	20.8	18.2	17.5	18.9	20.5	22.1	22.9	23.7	(30)
(31)		5%	WB	19.9	20.1	19.7	18.5	16.4	14.4	13.4	14.1	14.7	16.2	17.7	18.6	(31)
(32)		5%	MCDWB	22.9	23.2	23.0	21.6	20.0	17.6	16.7	17.8	19.2	20.7	21.5	22.3	(32)
(33)		10%	WB	19.1	19.4	19.1	17.8	15.6	13.6	12.6	13.3	13.9	15.4	16.9	17.9	(33)
(34)		10%	MCDWB	22.1	22.3	22.2	20.9	19.2	16.8	15.9	16.9	18.2	19.4	20.4	21.2	(34)
(35)	Mean Daily Temperature Range		MDBR	7.3	7.5	8.7	9.4	9.6	9.2	8.7	9.5	10.1	9.4	8.5	7.8	(35)
(36)		5% DB	MCDBR	10.8	11.7	14.6	14.2	13.3	11.7	11.1	13.1	14.6	15.8	14.1	11.8	(36)
(37)			MCWBR	5.5	5.9	6.4	6.6	6.5	6.4	6.1	6.6	7.0	7.1	6.4	6.0	(37)
(38)			5% WB	MCDWB	9.1	8.9	9.8	9.9	10.1	9.2	9.1	10.6	12.2	11.9	10.6	10.1
(39)		MCWBR		5.1	4.9	5.1	5.8	6.1	6.0	5.6	6.3	6.8	6.4	5.6	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.357	0.352	0.336	0.323	0.308	0.299	0.295	0.305	0.326	0.334	0.340	0.343	(40)	
(41)		taud	2.546	2.573	2.606	2.638	2.639	2.638	2.651	2.612	2.537	2.542	2.550	2.563	(41)	
(42)		Ebn at Noon	980	965	945	904	866	849	873	909	939	972	990	997	(42)	
(43)		Edh at Noon	109	102	92	80	70	66	68	80	97	104	108	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.31	6.38	5.28	3.95	2.95	2.48	2.68	3.56	4.77	6.02	6.91	7.55	(44)	
(45)		RadStd	0.52	0.43	0.34	0.23	0.18	0.10	0.13	0.14	0.17	0.33	0.40	0.45	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (7 neighbors)	+0.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+44	(47)

Nomenclature: See separate page